

TradeCircle

Platform Documentation

Deployment Guide & Admin Operator Guide

Phase 6 — Bootstrap admin, Firebase wizard,
modern UI, user CRUD, Vercel hardening

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Table of Contents

Part I — Deployment Guide

1. Prerequisites
2. GitHub Setup
3. Firebase Project Setup
4. Cloudinary Setup
5. Environment Variables Reference
6. Local Development (with bootstrap admin)
7. Vercel Deployment
8. First-Time Admin Setup
9. Firestore Security Rules Deployment
10. Health Check & Diagnostics
11. Custom Domain
12. Post-Deployment Verification
13. Troubleshooting
14. Common Errors Cookbook

Part II — Admin Operator Guide

1. First-Time Setup
2. Logging In
3. Admin Dashboard Overview
4. Creating Additional Admins
5. User Management
6. Authentication & Session
7. Permissions System
8. Common Workflows
9. Maintenance Schedule
10. Troubleshooting
11. Bootstrap Admin Flow (Local Dev)
12. User CRUD (Create / Edit / Delete)
13. Login Media Configuration
14. Health & Diagnostics
15. Resetting Your Admin Account

Part I — Deployment Guide

TradeCircle — Deployment Guide

A complete, beginner-friendly, step-by-step guide to deploying the **TradeCircle** platform to **Vercel** with **Firebase** as the backend and **Cloudinary** for image hosting.

If this is your first time deploying a Next.js app, follow each section in order. No step is optional unless explicitly marked.

Table of Contents

1. Prerequisites
2. GitHub Setup
3. Firebase Project Setup
4. Cloudinary Setup
5. Environment Variables Reference
6. Local Development (with bootstrap admin)
7. Vercel Deployment
8. First-Time Admin Setup
9. Firestore Security Rules Deployment
10. Health Check & Diagnostics
11. Custom Domain
12. Post-Deployment Verification
13. Troubleshooting
14. Common Errors Cookbook

1. Prerequisites

Before you begin, make sure you have:

Requirement	Purpose	Link
Node.js 20+	Runs the dev server and builds	https://nodejs.org/
Git	Version control	https://git-scm.com/
GitHub account	Hosts your code	https://github.com/
Vercel account (free tier OK)	Hosts the deployed app	https://vercel.com/signup
Firebase account (free Spark plan OK)	Auth, Firestore database, Storage, FCM push	https://firebase.google.com/
Cloudinary account (free tier OK)	Image uploads and CDN	https://cloudinary.com/users/register_free
A text editor (VS Code recommended)	Editing env files and code	https://code.visualstudio.com/

Verify Node and Git are installed:

```
node --version    # should print v20.x.x or higher
git --version     # should print git version 2.x.x
```

***Windows users:** PowerShell 5.1+ is the default shell in the documentation examples. Where bash equivalents differ, both are shown.*

2. GitHub Setup

If your project is not already in a Git repository:

```
# From the project root
git init
git add .
git commit -m "Initial commit"
```

Create a new **empty** repository on GitHub (do NOT add a README, .gitignore or license — your local repo already has them). Then:

```
git branch -M main
git remote add origin git@github.com:YOUR_USERNAME/tradecircle.git
git push -u origin main
```

Tip: If you use HTTPS instead of SSH, use
`https://github.com/YOUR_USERNAME/tradecircle.git`.

2.1 .gitignore essentials

Before your first push, confirm `.gitignore` includes:

```
.env
.env.local
.env.*.local
.tradecircle-local/
.next/
node_modules/
*.log
```

The `.tradecircle-local/` folder holds your **bootstrap admin credentials** (see Section 8). It **MUST** never be committed.

3. Firebase Project Setup

This is the most important external setup step. Get it wrong and you'll see cryptic 5xx errors in production. Follow each substep carefully.

3.1 Create the Firebase project

1. Go to `https://console.firebase.google.com/`
2. Click **Add project**, give it a name (e.g. `tradecircle-prod`), accept terms.
3. Disable Google Analytics for a simpler setup (you can always add GA4 later via the admin panel).
4. Wait ~30 seconds for the project to be created, then click **Continue**.

3.2 Enable Authentication (REQUIRED)

This is required before any user — including your first admin — can sign in.

1. Left sidebar: **Build** → **Authentication** → **Get started**.
2. On the **Sign-in method** tab, click **Email/Password** and toggle **Enable** → **Save**.

3. (Optional) Also enable **Google** — set a project support email, then Save.
4. (Optional) **Phone** — note: requires Blaze billing for production SMS; OK to skip for testing.

Without Email/Password enabled, the bootstrap admin migration step will fail with auth/operation-not-allowed.

3.3 Enable Cloud Firestore (REQUIRED)

1. Left sidebar: **Build** → **Firestore Database** → **Create database**.
2. Pick a region close to your users:
 - **australia-southeast1** (Sydney) — recommended for AU/NZ
 - **us-central1** — North America default
 - **europa-west3** (Frankfurt) — EU

The region cannot be changed later. Choose carefully.

1. Start in **production mode** (we will deploy security rules in Section 9).
2. Wait for the database to provision (~30 seconds).

Without Firestore created, all `/api/admin/*` routes will return 5 NOT_FOUND.

3.4 Enable Cloud Storage (REQUIRED)

1. Left sidebar: **Build** → **Storage** → **Get started**.
2. Accept the default location (must match your Firestore region).
3. Start in **production mode**.

Note: TradeCircle uses Cloudinary by default for image uploads, but Firebase Storage is still required by the SDK initialization.

3.5 Generate the Web SDK config

1. Project home → click the gear icon → **Project settings** → **General**.
2. Scroll to **Your apps** → click the `</>` Web icon.
3. App nickname: TradeCircle Web. **Do NOT** check "Also set up Firebase Hosting".
4. Click **Register app**. Firebase will display a config object — keep this tab open:

```
const firebaseConfig = {
  apiKey: "AIza...",
  authDomain: "tradecircle-prod.firebaseio.com",
  projectId: "tradecircle-prod",
  storageBucket: "tradecircle-prod.firebaseio.com",
  messagingSenderId: "123456789",
  appId: "1:123456789:web:abc123"
};
```

These six values become the `NEXT_PUBLIC_FIREBASE_*` env vars in Section 5.

3.6 Generate a Service Account JSON (REQUIRED)

This is a **separate download** from the Web SDK config above.

1. Project Settings → **Service Accounts** tab.
2. Click **Generate new private key** → confirm **Generate key**.
3. A JSON file downloads (e.g. tradecircle-prod-firebase-adminsdk-xxx.json).
4. Open the JSON in a text editor — you'll paste its full contents into the `FIREBASE_SERVICE_ACCOUNT_JSON` env var in Section 5.

***NEVER** commit this JSON to Git. Treat it like a password. The service account is **required for**: - /setup on Vercel (writing the master admin directly to Auth + Firestore) - /api/admin/* server routes (user CRUD, exports, backups) - /api/session session-cookie verification*

3.7 Note the Web Push (VAPID) key — optional

1. Project settings → **Cloud Messaging** tab.
2. Scroll to **Web Push certificates**, click **Generate key pair**.
3. Copy the **Key pair** value — this is `NEXT_PUBLIC_FIREBASE_VAPID_KEY`.

The Sender ID at the top of this tab is your `NEXT_PUBLIC_FIREBASE_MESSAGING_SENDER_ID`.

3.8 Summary — what you should now have

- [] A Firebase project (Web SDK config — 6 values)
- [] Email/Password sign-in **enabled**
- [] Firestore database **created** (production mode)
- [] Cloud Storage **enabled**
- [] Service account JSON downloaded to your computer
- [] (Optional) VAPID Web Push key

4. Cloudinary Setup

4.1 Create the account

1. Sign up at https://cloudinary.com/users/register_free.
2. After verifying your email, you'll land on the dashboard.

4.2 Get your Cloud Name

On the dashboard, you'll see **Cloud name**: e.g. `dxyz12345`. Copy this — it goes in `NEXT_PUBLIC_CLOUDINARY_CLOUD_NAME`.

4.3 Create an Unsigned Upload Preset

1. Click the gear icon → **Settings** → **Upload** tab.
2. Scroll to **Upload presets**, click **Add upload preset**.
3. **Preset name**: `tradecircle_unsigned` (or anything memorable).
4. **Signing mode**: change from "Signed" to **Unsigned**.
5. (Optional) Set a folder like `tradecircle/` so all uploads are grouped.
6. Click **Save**.

The preset name goes in `NEXT_PUBLIC_CLOUDINARY_UPLOAD_PRESET`.

If uploads fail with a 401 from Cloudinary, the preset is still Signed — re-check step 4.

5. Environment Variables Reference

Every variable the app reads, with description, example, where to get it, and which environments it belongs in.

Variable	Required?	Description	Example	Where to get	Environments
NEXT_PUBLIC_FIREBASE_API_KEY	Yes	Public web SDK API key	AIzaSyA...	Section 3.5	Prod / Preview / Dev
NEXT_PUBLIC_FIREBASE_AUTH_DOMAIN	Yes	OAuth redirect domain	tradecircle-prod.firebaseio.com	Section 3.5	Prod / Preview / Dev
NEXT_PUBLIC_FIREBASE_PROJECT_ID	Yes	Firestore project ID	tradecircle-prod	Section 3.5	Prod / Preview / Dev
NEXT_PUBLIC_FIREBASE_STORAGE_BUCKET	Yes	Cloud Storage bucket	tradecircle-prod.firebaseio.com	Section 3.5	Prod / Preview / Dev
NEXT_PUBLIC_FIREBASE_APP_ID	Yes	Web app identifier	1:123:web:abc	Section 3.5	Prod / Preview / Dev
NEXT_PUBLIC_FIREBASE_MESSAGING_SENDER_ID	Yes	FCM sender ID	123456789	Section 3.5 or 3.7	Prod / Preview / Dev
NEXT_PUBLIC_FIREBASE_VAPID_KEY	No	Web push VAPID key	BL...	Section 3.7	Prod / Preview / Dev
FIREBASE_SERVICE_ACCOUNT_JSON	Yes	Full service-account JSON (one line)	{"type": "service_account", ...}	Section 3.6	Prod / Preview / Dev
NEXT_PUBLIC_CLOUDINARY_CLOUD_NAME	Yes	Cloudinary cloud name	dxyz12345	Section 4.2	Prod / Preview / Dev
NEXT_PUBLIC_CLOUDINARY_UPLOAD_PRESET	Yes	Unsigned preset name	tradecircle_unsigned	Section 4.3	Prod / Preview / Dev
OPENAI_API_KEY	No	AI fraud detection, recs	sk-...	platform.openai.com	Prod (prefer Admin UI)
EMAILJS_PUBLIC_KEY	No	Transactional email	xxxxxxx	emailjs.com	Prod
EMAILJS_SERVICE_ID	No	EmailJS service ID	service_xxx	emailjs.com	Prod
NEXT_PUBLIC_STRIPE_PUBLISHABLE_KEY	No	Stripe public key	pk_live_...	dashboard.stripe.com	Prod
STRIPE_PAYMENT_LINK_URL	No	Stripe payment link	https://buy.stripe.com/...	Stripe Payment Links	Prod
EWAY_API_KEY	No	eWAY gateway	xxxxxx	eWAY portal	Prod
EWAY_PASSWORD	No	eWAY API password	xxxxxx	eWAY portal	Prod

Variable	Required?	Description	Example	Where to get	Environments
AWS_ACCESS_KEY_ID	No	S3 alt storage	AKIA...	AWS IAM	Prod
AWS_SECRET_ACCESS_KEY	No	S3 alt storage	xxxxx	AWS IAM	Prod
AWS_REGION	No	S3 region	us-east-1	AWS	Prod
AWS_S3_BUCKET	No	S3 bucket	tradecircle-uploads	AWS	Prod
NEXT_PUBLIC_GA_MEASUREMENT_ID	No	GA4 measurement ID	G-XXXXXXXXXX	analytics.google.com	Prod

5.1 Pasting the service-account JSON

When pasting `FIREBASE_SERVICE_ACCOUNT_JSON` into Vercel's env var UI, you can paste the multi-line JSON directly into the value field — Vercel preserves it correctly.

For your local `.env.local`, you must keep it on a **single line** (the embedded `\n` sequences inside `private_key` are already escaped — leave them as-is):

```
FIREBASE_SERVICE_ACCOUNT_JSON={"type":"service_account","project_id":"...", "private_key":"-----BEGIN PRIVATE KEY-----
```

6. Local Development (with bootstrap admin)

```
# 1. Install dependencies
npm install

# 2. Create your local env file
cp .env.example .env.local      # PowerShell: Copy-Item .env.example .env.local

# 3. Open .env.local and paste the values from Section 5
#   (Firebase config is optional for first-run – see below)

# 4. Start the dev server
npm run dev
```

Open `http://localhost:3000` — the homepage should load.

6.1 Two ways to run locally

TradeCircle supports two local-dev modes:

1. **Full Firebase mode** — `.env.local` has every `NEXT_PUBLIC_FIREBASE_*` plus `FIREBASE_SERVICE_ACCOUNT_JSON`. The `/setup` page writes directly to Firebase Auth + Firestore.
1. **Bootstrap mode** — `.env.local` is empty (or missing Firebase keys). The `/setup` page writes the admin to a local JSON file at `.tradecircle-local/config.json`. You can later run the **Firebase Setup Wizard** at `/admin/firebase-setup` to migrate to a real Firebase project.

Bootstrap mode lets developers explore the platform without needing a Firebase project on day one.

6.2 Clearing dev caches

If you've just installed new dependencies or upgraded Next.js and see stale errors, clear the `.next` cache:

```
# PowerShell
Remove-Item -Recurse -Force .next
npm run dev

# bash / zsh
```

```
rm -rf .next  
npm run dev
```

7. Vercel Deployment

7.1 Import the project

1. Go to <https://vercel.com/new>.
2. Click **Add New Project** → **Import Git Repository** and select your TradeCircle repo.
3. Vercel auto-detects **Next.js**. Leave defaults:
 - **Framework Preset:** Next.js
 - **Build Command:** `npm run build` (do not change)
 - **Output Directory:** `.next`
 - **Install Command:** `npm install`
 - **Node.js Version:** 20.x (the project requires 20+)

7.2 Add environment variables — BEFORE first deploy

Click **Environment Variables** and add every variable from Section 5. **Add them before clicking Deploy** — a half-configured first deployment frequently caches missing-env warnings.

For each variable:

- **Name:** e.g. `NEXT_PUBLIC_FIREBASE_API_KEY`
- **Value:** paste the actual value
- **Environments:** tick **Production**, **Preview**, and **Development** (all three)

For `FIREBASE_SERVICE_ACCOUNT_JSON`: open the downloaded JSON file in a text editor, select all (Ctrl+A), copy (Ctrl+C), and paste the entire content into the value field. Vercel handles the multi-line value correctly.

7.3 Deploy

Click **Deploy**. The first build takes 2–5 minutes. When it finishes you'll get a URL like `https://tradecircle-xyz.vercel.app`.

Visit the URL. The homepage should load. If it shows "Page couldn't load", jump to Section 10 (health check) before anything else.

7.4 Subsequent deploys

Every push to `main` triggers a new production deploy automatically. Pushes to any other branch create a Preview deployment.

To redeploy without a code change (e.g. after adding an env var):

- Vercel → Deployments → click the  menu → **Redeploy**.

8. First-Time Admin Setup

TradeCircle uses a **hybrid bootstrap** model so the platform is usable both with and without Firebase fully configured.

8.1 Hybrid behavior — what happens at `/setup`

Environment	What <code>setup</code> does
Vercel (production)	Detects <code>VERCEL=1</code> . Uses Firebase Admin SDK to write directly to Firebase Auth + Firestore. Requires <code>FIREBASE_SERVICE_ACCOUNT_JSON</code> . Never touches the filesystem.
Local dev (full Firebase)	If <code>.env.local</code> has Firebase keys, writes to Firebase Auth + Firestore.
Local dev (bootstrap)	If no Firebase keys, writes admin credentials to <code>.tradecircle-local/config.json</code> . You can migrate later.

The local-file path is **disabled on Vercel** to avoid `ENOENT: no such file or directory, mkdir /var/task/.tradecircle-local`. Environment detection (`process.env.VERCEL`) gates this.

8.2 Walk-through — the `/setup` form

- Visit `https://yourdomain.com/setup` (or `http://localhost:3000/setup`).
- Fill in:
 - Full Name** — e.g. *Jane Doe*
 - Email** — admin email (e.g. *admin@yourdomain.com*)
 - Password** — min 8 chars, 1 uppercase, 1 number
 - Confirm Password** — must match
- Click **Create Master Admin**.
- You'll be redirected to `/admin/dashboard`.

The page locks itself after first use — subsequent admins are added via the admin panel.

8.3 Walk-through — the Firebase Setup Wizard (local dev only)

If you started in bootstrap mode and now want to connect to a real Firebase project, visit `/admin/firebase-setup`. The wizard has four steps:

Step	Title	What it asks for
1	Welcome	Overview, links to Section 3 of this guide
2	Keys + Test	The six <code>NEXT_PUBLIC_FIREBASE_*</code> values. Click Test connection to verify with a real Firebase API call.
3	Service Account	Paste the contents of the service account JSON. Validated client-side.
4	Download <code>.env</code>	Generates a complete <code>.env.local</code> for you to download. Restart <code>npm run dev</code> to apply.

8.4 Migrate bootstrap admin to Firebase

If you created your admin in bootstrap mode and want to move them into Firebase:

- Complete the Firebase Setup Wizard (8.3) and restart the dev server.
- Visit `/admin/firebase-setup` again.
- A new banner **"Migrate Admin to Firebase"** is displayed.

4. Click it. The system:

- Creates the user in Firebase Auth using the bootstrap email + a temporary password
- Writes the `users/{uid}` document with role `super-admin`
- Sends a password reset email so the admin can choose their real password
- Deletes the entry from `.tradecircle-local/config.json`

8.5 Resetting the local admin

If you forget the bootstrap password or want to start over:

```
curl -X DELETE http://localhost:3000/api/local-auth/reset
```

Or simply delete the file:

```
Remove-Item .tradecircle-local\config.json
```

Then revisit `/setup`.

9. Firestore Security Rules Deployment

The `firestore.rules` file in the repo defines who can read/write each Firestore collection. **These must be deployed** — without them, every Firestore call fails with `Missing or insufficient permissions`.

9.1 Method A — Paste into Firebase Console (no CLI)

1. Open `firestore.rules` from the repo in your editor and **copy the entire file**.
2. Go to Firebase Console → **Firestore Database** → **Rules**.
3. Select all in the editor (Ctrl+A) and paste your rules over the default.
4. Click **Publish**.
5. Wait for the green confirmation banner.

Alternatively, link directly to the raw file on GitHub:

```
https://raw.githubusercontent.com/YOUR_USERNAME/tradecircle/main/firestore.rules
```

Then copy/paste into the console.

9.2 Method B — Firebase CLI

```
npm install -g firebase-tools
firebase login
firebase use --add          # pick your project
firebase deploy --only firestore:rules
```

9.3 When to redeploy rules

Redeploy any time `firestore.rules` changes — typically after pulling new code that touches access patterns (new collection, changed role checks, etc.).

9.4 What the rules allow (summary)

- `users/{uid}` — readable by the owner and any admin; writable by the owner or an admin
- `products/*` — public read; write restricted to the seller (`ownerId == request.auth.uid`)
- `messages/*` — readable/writable only by participants
- `adminLogs/*` — admin-only read; server-side write
- See `firestore.rules` for the authoritative list

10. Health Check & Diagnostics

The platform ships a built-in health endpoint at `/api/health`. It's the **first thing to check** if a deploy looks broken.

10.1 Hitting the endpoint

```
curl https://your-app.vercel.app/api/health
```

Sample response:

```
{
  "ok": true,
  "firebase": true,
  "firebaseAdmin": true,
  "cloudinary": true,
  "timestamp": "2026-05-17T12:34:56.789Z"
}
```

10.2 What each boolean means

Field	True when...	If false, fix by...
firebase	All six NEXT_PUBLIC_FIREBASE_* env vars are present	Add the missing client env vars in Vercel, redeploy
firebaseAdmin	FIREBASE_SERVICE_ACCOUNT_JSON parses successfully	Re-paste the service account JSON (full content, valid JSON), redeploy
cloudinary	Both cloud name + upload preset env vars are set	Add Cloudinary env vars in Vercel, redeploy
ok	All of the above are true	See specific field that is false

10.3 Diagnostic workflow

1. Open `/api/health` first. If any field is `false`, fix the env var before debugging anything else.
2. If everything is `true` but pages still fail, check **Vercel** → **Deployments** → **latest** → **Functions** for runtime errors.
3. If Firestore reads fail, jump to Section 9 (rules deployment).
4. If Auth login fails, confirm Email/Password is enabled in Firebase (Section 3.2).

11. Custom Domain

11.1 Add domain in Vercel

1. Vercel project → **Settings** → **Domains**.
2. Type your domain (e.g. `tradecircle.com.au`), click **Add**.
3. Vercel shows DNS records to configure.

11.2 Configure DNS

At your registrar (GoDaddy, Cloudflare, Namecheap, etc.):

- **Root domain** (`tradecircle.com.au`): A record → `76.76.21.21`
- **www subdomain**: CNAME → `cname.vercel-dns.com`

Save. DNS propagation usually takes 1–30 minutes.

11.3 SSL

Vercel auto-provisions a Let's Encrypt SSL cert (typically <1 minute after DNS resolves). The domain badge turns green.

11.4 Update Firebase authorized domains

Firebase → **Authentication** → **Settings** → **Authorized domains** → **Add domain** → enter `tradecircle.com.au` (and `www.tradecircle.com.au` if used).

Without this, Google Sign-In and password reset emails fail on your production domain.

12. Post-Deployment Verification

Run through this checklist after the first successful deploy.

- `[] /api/health` returns all-true
- `[]` Homepage loads at your custom domain
- `[] /setup` creates the master admin (verify in Firebase → Authentication → Users)
- `[] /login` signs in with that admin
- `[] /admin/dashboard` is accessible and shows zero errors in the browser console
- `[] /admin/users` lists the master admin
- `[]` Create a test user from `/admin/users` → **Add User** — receives invite email
- `[] /products/new` — upload an image (verify in Cloudinary → Media Library)
- `[] /messages` — send a message; appears in real-time
- `[] /settings` — push permission prompt appears (if VAPID configured)
- `[]` No "Missing Firebase env var" warnings in browser console
- `[]` Vercel → Deployments → Functions tab: no 500 errors on `/api/session` or `/api/admin/*`

13. Troubleshooting

"Page couldn't load" / blank screen on Vercel

Missing env vars are the #1 cause. **First step:** hit `/api/health` and check which booleans are `false`. Fix those vars in Vercel → Settings → Environment Variables, then redeploy.

5 NOT_FOUND in Vercel logs

Firestore database was not created (Section 3.3). Go to Firebase Console → Firestore Database → **Create database**.

CONFIGURATION_NOT_FOUND (Firebase Auth)

The Email/Password sign-in provider is not enabled. Firebase Console → Authentication → Sign-in method → **Email/Password** → **Enable**.

Missing or insufficient permissions

`firestore.rules` has not been deployed. Follow Section 9.

ENOENT: no such file or directory, mkdir `/var/task/.tradecircle-local`

A local-file write code path is being called on Vercel. This was **fixed** by the `process.env.VERCEL` environment guard in `/api/setup` and `/api/local-auth/*`. If you see this on a fresh deploy, pull the

latest main branch.

JSON.parse: unexpected end of data

A server route returned an empty response body. Open **Vercel** → **Deployments** → **Functions** for the route in question and read the runtime logs — there's usually a thrown error before the response is written.

auth/email-already-exists on /setup

A master admin already exists for this email. Either:

- Use the **Forgot password** link at `/login`, or
- Delete the user from Firebase Console → Authentication → Users → ■ → Delete, then revisit `/setup`.

"Incorrect email or password" on /login

- The email or password doesn't match what's stored in Firebase Auth.
- Use the **Forgot password** link to reset, or reset the password from Firebase Console → Authentication → Users → ■ → Reset password.

Stale .next cache after npm install

```
Remove-Item -Recurse -Force .next  
npm run dev
```

Vercel build fails on /following prerender

Fixed by wrapping the page in a server component with `export const dynamic = 'force-dynamic'`. If you still see it, ensure `app/following/page.tsx` exports `dynamic = 'force-dynamic'` and that any client logic lives in a separate child component.

Browser error: applyTheme is not a function

Fixed by exposing `setTheme` as an alias on the theme provider context. If a custom theme component still throws, ensure it's importing from the project's `ThemeProvider` and not a stale module.

Tailwind utilities not applying (especially max-w-*)

Fixed in `app/globals.css` with `@utility` overrides using `!important` for the t-shirt sizes. If you add a new utility and it's overridden by another rule, add it to the override block.

Google Sign-In returns auth/unauthorized-domain

Your production domain is not in Firebase's authorized domain list. Firebase → Authentication → Settings → Authorized domains → Add domain.

Phone auth doesn't work

Phone auth requires the Firebase **Blaze (pay-as-you-go)** plan in production. Upgrade in Firebase Console → Usage & billing → Modify plan.

Push notifications never arrive

- VAPID key missing or wrong.
- Service worker not registered (DevTools → Application → Service Workers).
- Confirm `NEXT_PUBLIC_FIREBASE_VAPID_KEY` is set, hard-refresh, accept the browser permission prompt.

"Quota exceeded" on Firestore reads

You've hit the Spark plan's daily quota (50K reads). Upgrade to Blaze — still effectively free for small apps.

Build fails on Vercel with TypeScript errors

`next.config.ts` ships with `typescript.ignoreBuildErrors: true` already configured. If you still see errors, confirm:

```
typescript: { ignoreBuildErrors: true },
eslint: { ignoreDuringBuilds: true },
```

14. Common Errors Cookbook

A one-stop table of every error message we've seen during Phase 6 deployments, with root cause and fix.

Error message	Root cause	Fix	Reference
Page couldn't load (blank)	Missing env vars on Vercel	Check <code>/api/health</code> and add missing vars	\$10
5 NOT_FOUND	Firestore DB not created	Create Firestore in production mode	\$3.3
CONFIGURATION_NOT_FOUND	Email/Password sign-in disabled	Enable in Firebase Auth → Sign-in method	\$3.2
Missing or insufficient permissions	Firestore rules not deployed	Paste <code>firestore.rules</code> into Firebase Console	\$9
ENOENT ... /var/task/.tradecircle-local	Local-file path called on Vercel	Pull latest main (env detection added)	\$13
JSON.parse: unexpected end of data	Server route returned empty body	Read Vercel function logs	\$13
auth/email-already-exists	Admin exists already	Reset password or delete in Firebase Console	\$13
Incorrect email or password	Password mismatch	Forgot Password link, or reset in console	\$13
auth/unauthorized-domain	Domain not authorized in Firebase	Add to Authorized domains	\$11.4
auth/operation-not-allowed	Sign-in provider disabled	Enable Email/Password	\$3.2
applyTheme is not a function	Stale theme provider	Pull latest main (setTheme alias added)	\$13
Tailwind max-w-* not applying	Utility override conflict	Already fixed in <code>globals.css@utility</code> block	\$13
Cloudinary 401 on upload	Preset still Signed	Change preset to Unsigned	\$4.3
firebase-admin/app initialization failure	FIREBASE_SERVICE_ACCOUNT_JSON malformed	Re-paste full JSON contents	\$3.6
/following prerender failure	Static prerender on dynamic page	Confirm <code>dynamic = 'force-dynamic'</code>	\$13
Stale errors after dep install	<code>.next</code> cache	<code>Remove-Item -Recurse -Force .next</code>	\$6.2

Error message	Root cause	Fix	Reference
Push notifications missing	VAPID key missing / SW not registered	Set VAPID, hard refresh, allow prompt	§13
Firestore "Quota exceeded"	Spark free tier limit	Upgrade to Blaze	§13

Need more help? Check the project README for support links, or open an issue on GitHub.

Part II — Admin Operator Guide

TradeCircle — Admin Guide

A friendly guide to administering your TradeCircle platform. Written for non-technical operators — no coding required.

Looking to deploy the app? See `DEPLOYMENT.md` instead.

Table of Contents

1. First-Time Setup
2. Logging In
3. Admin Dashboard Overview
4. Creating Additional Admins
5. User Management
6. Authentication & Session
7. Permissions System
8. Common Workflows
9. Maintenance Schedule
10. Troubleshooting
11. Bootstrap Admin Flow (Local Dev)
12. User CRUD (Create / Edit / Delete)
13. Login Media Configuration
14. Health & Diagnostics
15. Resetting Your Admin Account

1. First-Time Setup

Right after the app is deployed for the first time, you need to create the master administrator account.

1. Open your domain in a browser: `https://yourdomain.com/setup`
2. Fill in the form:
 - **Full Name** — e.g. *Jane Doe*
 - **Email** — your admin email (e.g. *admin@yourdomain.com*)
 - **Password** — minimum 8 characters, at least 1 uppercase letter and 1 number
 - **Confirm Password** — same as above
1. Click **Create Master Admin**.
2. You'll be redirected to `/admin/dashboard`.

*The `/setup` page **locks itself permanently** after the first super-admin is created. All additional admins must be added from inside the admin panel (see Section 4).*

2. Logging In

1. Visit `https://yourdomain.com/login`.
2. Enter your admin email and password.
3. After signing in, click your avatar (top right). If you're an admin, you'll see an **Admin Dashboard** link.
4. Click it to enter `/admin/dashboard`.

*If you forget your password, use the **Forgot password?** link on the login page. A reset email will be sent.*

3. Admin Dashboard Overview

The admin panel is organized as a left-hand sidebar. Each entry below opens a dedicated page.

Section	What it does
Dashboard	High-level stats: total users, active listings, GMV, recent activity feed.
Users	Manage buyers, sellers and advisors — search, view, ban, unban, delete.
Admin Users	Manage administrator accounts — create new admins, assign roles, toggle module permissions.
Fraud Flags	Review listings/users flagged by AI for suspicious activity. Dismiss or take action.
Products	Moderate marketplace listings — feature, hide, remove, bulk-edit.
Advisories	Moderate advisor posts and articles.
Enquiries	View incoming contact-form submissions and respond.
Orders	Order history, payment status, refunds, fulfillment notes.
Ads	Manage banner ads and sponsored content placement.
Ad Campaigns	A/B test ad creatives and track conversions.
Marketing SSO	Connect Mailchimp / HubSpot / Meta accounts for marketing automation.
Loyalty Program	Configure point earning rules, tiers, and reward catalogue.
AI Settings	Configure OpenAI (or alternative) API key, model, and feature toggles.
CMS	Edit blog posts, FAQ entries, and static pages (About, Terms, Privacy).
Tax	Define regional tax rates (GST/VAT/sales tax).
Feature Toggles	Turn entire modules (chat, advisors, loyalty, etc.) ON/OFF without redeploying.
Configuration	Branding (logo/colors), backend selection (Firebase vs. Supabase), payment gateways, currency, theme.
Tenant Branding	White-label settings for multi-tenant deployments.
Operator Billing	Manage subscription tiers for tenants/operators.

Section	What it does
Operators	List and manage multi-tenant operator accounts.
Storage	Switch active storage backend between Cloudinary and AWS S3.
Analytics	KPIs, charts, top sellers, GMV trends.
Integrations	GA4 measurement ID, Facebook Pixel ID, other tracking.
Exports	Download CSV reports (users, orders, products, transactions).
System Backup	Create, restore, and schedule Firestore backups.
Roles	Define regional admin hierarchies (country/region managers).
API Keys	Generate and revoke API keys for third-party integrations.
System Status	Service Level Agreement (SLA), incident log, current health.
Geo-Targeted Ads	Location-based ad delivery rules.
Data Residency	Choose Firebase region for compliance (e.g. EU, AU).

4. Creating Additional Admins

1. From the sidebar, click **Admin Users**.
2. Click **Add Admin** (top right).
3. Fill in:
 - **Name** — the new admin's full name
 - **Email** — their work email (login email)
 - **Role** — pick one of:
 - **super-admin** — full access to everything
 - **admin** — broad access, but cannot manage other admins
 - **moderator** — limited to user/content moderation
 - **analyst** — read-only access to analytics & exports
1. Toggle individual **module permissions** on/off as needed.
2. Click **Save**.

The new admin will receive a **password reset email**. They click the link, set their own password, and can then log in at `/login`.

You can change roles or permissions any time by clicking the admin in the list and editing.

5. User Management

1. Sidebar → **Users**.
2. Use the search bar at the top — search by email, name, or user ID.
3. Click a row to expand the user's detail panel.
4. Available actions:

- **Ban** — sets `active = false`. User cannot log in. Listings hidden.
- **Unban** — restores access.
- **Delete** — permanent. Wipes user record and revokes auth.
- **View Activity** — orders, messages, listings, login history.

*Banning is recoverable. Deletion is **not**. When in doubt, ban first.*

6. Authentication & Session

The platform uses HTTP-only secure cookies to maintain logged-in sessions:

Cookie	Purpose
tc-session	Encrypted ID token (validates user identity on every request)
tc-role	User role (buyer/seller/advisor/admin) for fast UI rendering
tc-admin	Admin flag for gated admin routes

- **Session duration**: 7 days. After that, the user must log in again.
- **Logout**: clicking Logout clears all three cookies and revokes the refresh token on Firebase, immediately ending the session on all devices.

7. Permissions System

Each admin has **13 module-level permission flags**. They control which sidebar entries are visible to that admin.

- **super-admin** — all 13 flags automatically `true`. Cannot be reduced. Full power.
- **admin** — preset with most flags `true` except admin-user-management.
- **moderator** — preset focused on Users, Products, Advisories, Fraud Flags, Enquiries.
- **analyst** — preset focused on Analytics, Exports (read-only).

You can override any preset by toggling individual permissions in the Admin Users edit dialog. Saving instantly updates that admin's access (they may need to refresh).

8. Common Workflows

Workflow 1 — Approve or reject a flagged product

1. Sidebar → **Fraud Flags**.
2. Browse the list (most recent first).
3. Click a row to open details — see the flag reason, AI confidence score, and the listing preview.
4. Decide:
 - **Dismiss flag** — listing stays live, flag cleared.
 - **Remove listing** — listing is hidden, seller notified.
 - **Ban seller** — escalate if the seller has multiple flags.

Workflow 2 — Ban a problem user

1. Sidebar → **Users**.
2. Search by email or name.
3. Click their row.
4. Click **Ban** → confirm. The user is signed out immediately on all devices.

Workflow 3 — Update site branding

1. Sidebar → **Configuration**.
2. Open the **Branding** tab.
3. Upload your logo (PNG or SVG), set primary/accent colors, set tagline.
4. Click **Save**. Changes go live within ~30 seconds.

Workflow 4 — Schedule a weekly Firestore backup

1. Sidebar → **System Backup**.
2. Open the **Auto Backup** tab.
3. Toggle **Enabled** on.
4. Frequency: **Weekly**, Day: **Sunday**, Time: **02:00**.
5. Retention: 8 weeks.
6. Click **Save**. The system will run the backup automatically.

Workflow 5 — Set up Google Analytics 4 tracking

1. Get your GA4 Measurement ID from <https://analytics.google.com/> (format: G-XXXXXXXXXX).
2. Sidebar → **Integrations**.
3. Paste the ID into the **GA4 Measurement ID** field.
4. Click **Save**. Pageviews start tracking within a few minutes.

9. Maintenance Schedule

A recommended cadence to keep the platform healthy:

Frequency	Tasks
Daily	Review Fraud Flags , respond to open Enquiries , scan System Status for incidents.
Weekly	Review Analytics trends, verify last Backup ran, check Orders for stuck payments.
Monthly	Review user growth, subscription/loyalty metrics, top sellers, and revenue.
Quarterly	Clean up archived data, rotate API keys , review admin user list (remove ex-employees), audit role permissions.

10. Troubleshooting

"Can't log in"

- Use **Forgot password?** on the login page to reset.
- Make sure the email is exactly the one you registered with.

- If still stuck, ask a super-admin to delete and re-create your admin record.

"Permission denied" when opening a sidebar item

- Your admin role doesn't include that module flag.
- Ask a super-admin to enable the module for you in **Admin Users** → your row → toggle the flag.

"Page won't load" / blank screen

- Check **System Status** for active incidents.
- Hard refresh the browser (Ctrl/Cmd + Shift + R).
- Try an Incognito window to rule out a broken cookie.
- Check the deployed app's status page in Vercel.

"Image won't upload"

- Verify in **Storage** that an upload backend is selected and credentials are filled in.
- Ensure your Cloudinary preset is **unsigned** (see `DEPLOYMENT.md` §4.3).

"AI features aren't working"

- Sidebar → **AI Settings** — verify the OpenAI API key is set and the toggle is on.
- Check that your OpenAI account has credit.

"User got an email I didn't intend to send"

- Sidebar → **CMS / Marketing SSO** — review automation rules.
- Disable any automation triggers you don't recognize.

Need more help? Contact your technical administrator or refer to `DEPLOYMENT.md` for infrastructure-level issues.

11. Bootstrap Admin Flow (Local Dev)

When you run TradeCircle locally without a Firebase project configured, the platform falls back to a **bootstrap mode** so you can still log in and explore. This section explains how it works and how to graduate to a real Firebase project.

11.1 What is `.tradecircle-local/`?

A hidden folder created in the project root the first time you submit `/setup` without Firebase env vars. It stores a single file:

```
.tradecircle-local/  
  ■■ config.json  # bootstrap admin email + hashed password + role
```

- The hashed password uses bcrypt with a per-install random salt.
- The role is always `super-admin`.
- The folder is created with `0700` permissions on POSIX systems.

11.2 Why is the folder gitignored?

`.tradecircle-local/` contains credentials. It is in `.gitignore` and must **never** be committed. On Vercel the folder is never created — production always uses Firebase.

11.3 Running the Firebase Setup Wizard

To graduate from bootstrap to a real Firebase project:

1. Log in as the bootstrap admin (`/login`).
2. Visit `/admin/firebase-setup`.
3. Step through the four pages:
 - **Welcome** — overview and link to `DEPLOYMENT.md §3`
 - **Keys + Test** — paste the six `NEXT_PUBLIC_FIREBASE_*` values, click **Test connection**
 - **Service Account** — paste the contents of the downloaded JSON
 - **Download .env** — click the button to download a complete `.env.local`
1. Replace your project's `.env.local` with the downloaded file.
2. Restart `npm run dev`.

11.4 Downloading the `.env` file

The wizard generates a fully-populated `.env.local` containing all six client vars plus the service-account JSON on a single line. Save it to the project root.

If you already have a `.env.local`, the downloaded file will overwrite it — back up first if needed.

11.5 Migrating the bootstrap admin to Firebase

Once Firebase is connected:

1. Visit `/admin/firebase-setup` again.
2. Click the **Migrate Admin to Firebase** banner.
3. The system creates the user in Firebase Auth, writes the `users/{uid}` doc with role `super-admin`, sends a password reset email, and deletes `.tradeircle-local/config.json`.
4. Log out, then log back in with the new password from the reset email.

12. User CRUD (Create / Edit / Delete)

`/admin/users` is the user-management page. Phase 6 added a modern modal-driven CRUD UI with bulk actions and audit logging.

12.1 Page tour

Element	Purpose
Search bar	Filter by email, name, or UID
Role filter	Buyer / Seller / Advisor / Admin
Status filter	Active / Banned / Pending
Date range	Created-on filter
Add User button (top right)	Opens the Create User modal
Row checkbox	Selects user for bulk action
Row click	Opens the Edit User modal
Bulk toolbar	Appears when at least one row is selected

12.2 Create User modal

Fields:

- **Name** — full name
- **Email** — login email (validated)
- **Role** — Buyer / Seller / Advisor / Admin / Super-admin
- **Initial password** (*optional*) — if provided, account is created active immediately
- **Send invite email** (*optional*) — alternative to password; sends a password-set link via Firebase

Click **Create**. The new user appears in the list within a second.

12.3 Edit User modal

Editable fields:

- Name
- Role (super-admins only)
- Active flag (Ban / Unban)
- Module permission toggles (for admin roles)

Saving fires `/api/admin/users/:uid` (PATCH) and writes an audit entry to `adminLogs`.

12.4 Bulk actions toolbar

Available actions when one or more rows are selected:

Action	Effect
Ban	Sets <code>active = false</code> on every selected user
Unban	Sets <code>active = true</code>
Delete	Permanent. Removes user record and revokes Firebase Auth account
Export CSV	Downloads selected rows as CSV (id, email, name, role, createdAt, active)

12.5 Audit log

Every create/edit/delete writes a record to the `adminLogs` Firestore collection:

```
adminLogs/{autoId}
■■ actorUid: string          # admin who performed the action
■■ actorEmail: string
■■ action: "user.create" | "user.update" | "user.delete" | ...
■■ targetUid: string
■■ before: object | null     # snapshot of the user pre-change
■■ after: object | null      # snapshot post-change
■■ at: timestamp
```

Review at `/admin/audit-log` (super-admin only).

13. Login Media Configuration

The `/admin/login-media` page controls the carousel of images/videos that animates behind the login form.

13.1 Adding a slide

1. Click **Add slide**.
2. Choose type: **Image** or **Video**.

3. Click the upload zone — the Cloudinary widget opens.
4. Pick a file from your computer or a previously-uploaded Cloudinary asset.
5. The slide appears in the list. Drag to reorder.

13.2 Cloudinary upload integration

The widget uses your configured `NEXT_PUBLIC_CLOUDINARY_CLOUD_NAME` and `NEXT_PUBLIC_CLOUDINARY_UPLOAD_PRESET`. Uploaded URLs are stored in the `loginMedia` Firestore document.

13.3 Blur and interval sliders

- **Blur (px)** — applies a CSS backdrop-filter blur to the background. Range 0–40, default 8.
- **Interval (seconds)** — how long each slide is shown before crossfading to the next. Range 3–30, default 6.

13.4 Live preview

A real-time preview at the bottom of the page shows exactly what the login screen will look like. Saving with **Publish** propagates the change instantly — no redeploy needed.

14. Health & Diagnostics

14.1 The `/api/health` endpoint

A lightweight JSON endpoint that reports whether the deploy is correctly configured.

```
curl https://yourdomain.com/api/health
```

```
{
  "ok": true,
  "firebase": true,
  "firebaseAdmin": true,
  "cloudinary": true,
  "timestamp": "2026-05-17T12:34:56.789Z"
}
```

14.2 What each boolean means

Field	True when...
firebase	All six <code>NEXT_PUBLIC_FIREBASE_*</code> env vars present
firebaseAdmin	<code>FIREBASE_SERVICE_ACCOUNT_JSON</code> parses successfully
cloudinary	Cloud name + upload preset env vars set
ok	All of the above true

14.3 Using it to verify Vercel env vars

After any env-var change in Vercel → redeploy → hit `/api/health`. Any `false` field tells you exactly which group of vars to re-check. See `DEPLOYMENT.md §10` for the full diagnostic workflow.

15. Resetting Your Admin Account

15.1 Local dev (bootstrap admin)

If you forgot your bootstrap password:

```
# Option 1 – API
curl -X DELETE http://localhost:3000/api/local-auth/reset
```

```
# Option 2 – delete the file
# PowerShell
Remove-Item .tradecircle-local\config.json
# bash
rm .tradecircle-local/config.json
```

Then visit `/setup` and create a new master admin.

15.2 Production (Firebase)

1. Easy path: use **Forgot password?** on the `/login` page. A reset email is sent.
2. Manual path: Firebase Console → **Authentication** → **Users** → find your email → click ■ → **Reset password** (sends an email) or **Delete account** (then revisit `/setup`).

15.3 Re-running `/setup`

After deleting your admin record (either method above):

1. Visit `https://yourdomain.com/setup` again.
2. The page is unlocked because no super-admin exists.
3. Fill in the form to create a fresh master admin.

Re-running `/setup` does NOT delete other users, products, or data. It only creates a new master admin.